

JLU

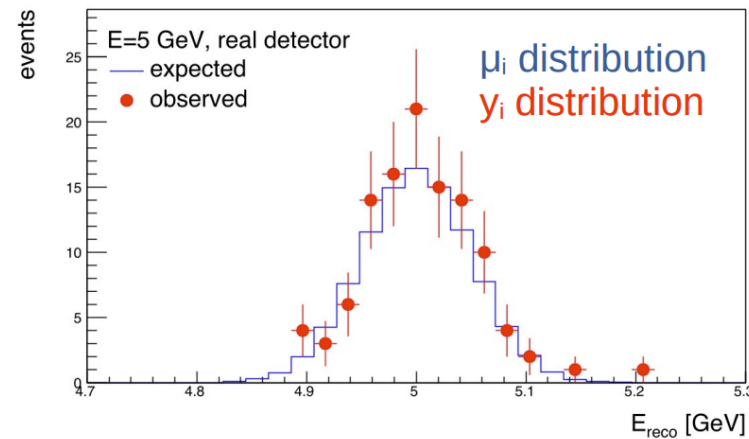
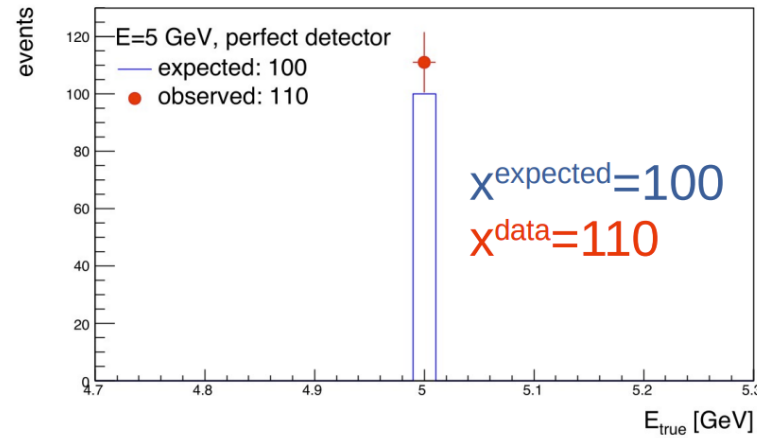
NEUE WEGE. SEIT 1607.

JUSTUS-LIEBIG-
 UNIVERSITÄT
GIESSEN

Detector Response Function & Data Unfolding Basics

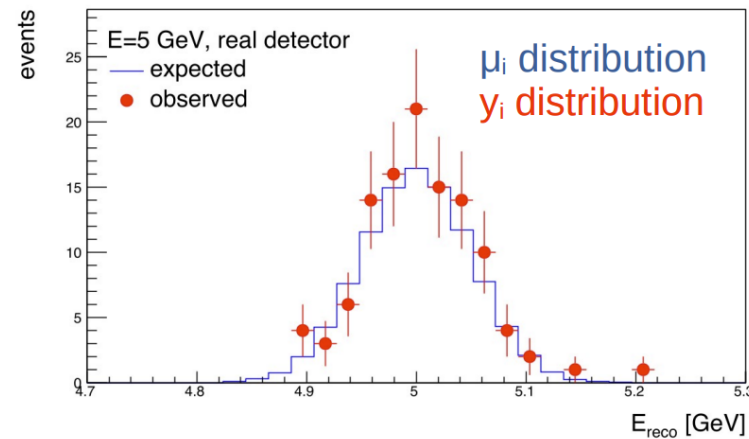
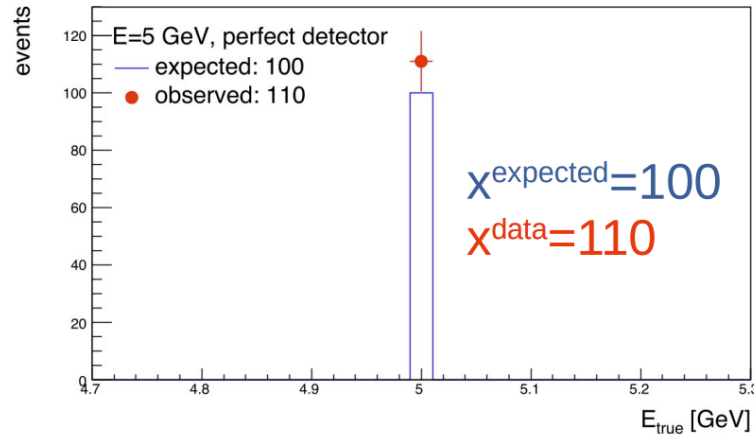
Background

- Real detectors distort observables during measurements
 - i.e. charge, amp etc.



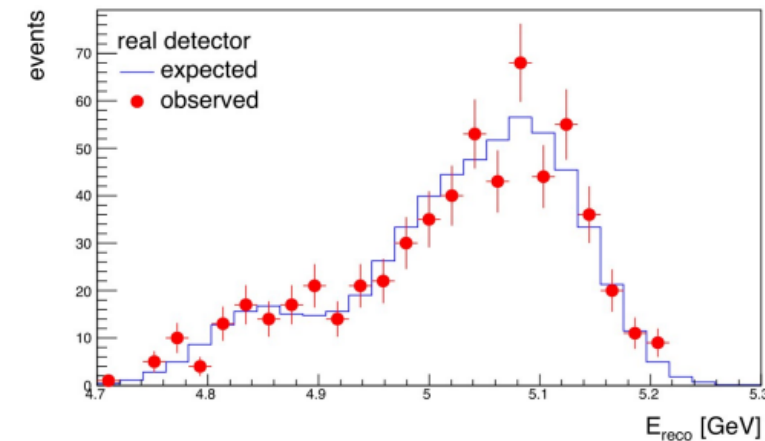
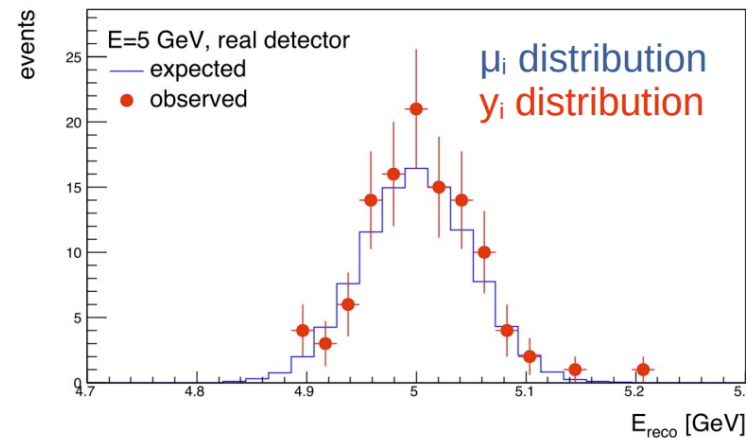
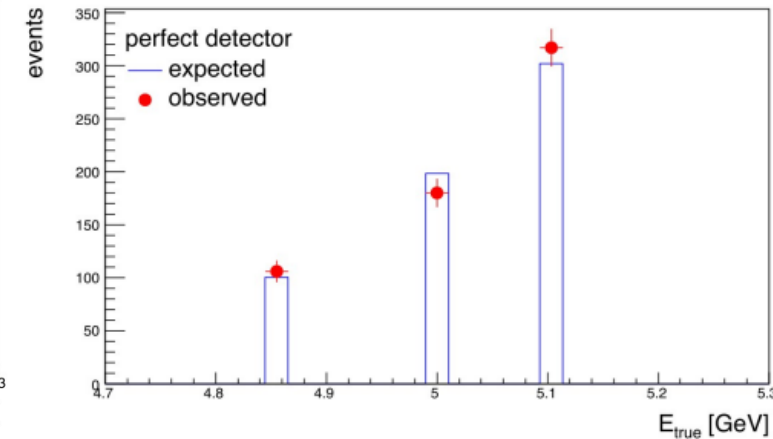
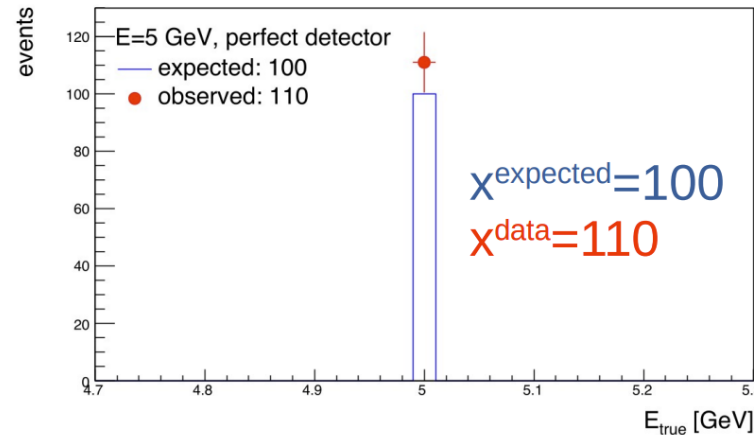
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Folding

$$n(E_i) = \sum_j R_{ij} \cdot n(C_j) + b_i$$

Effect (Measurement) Detector Response Matrix Cause (Truth)

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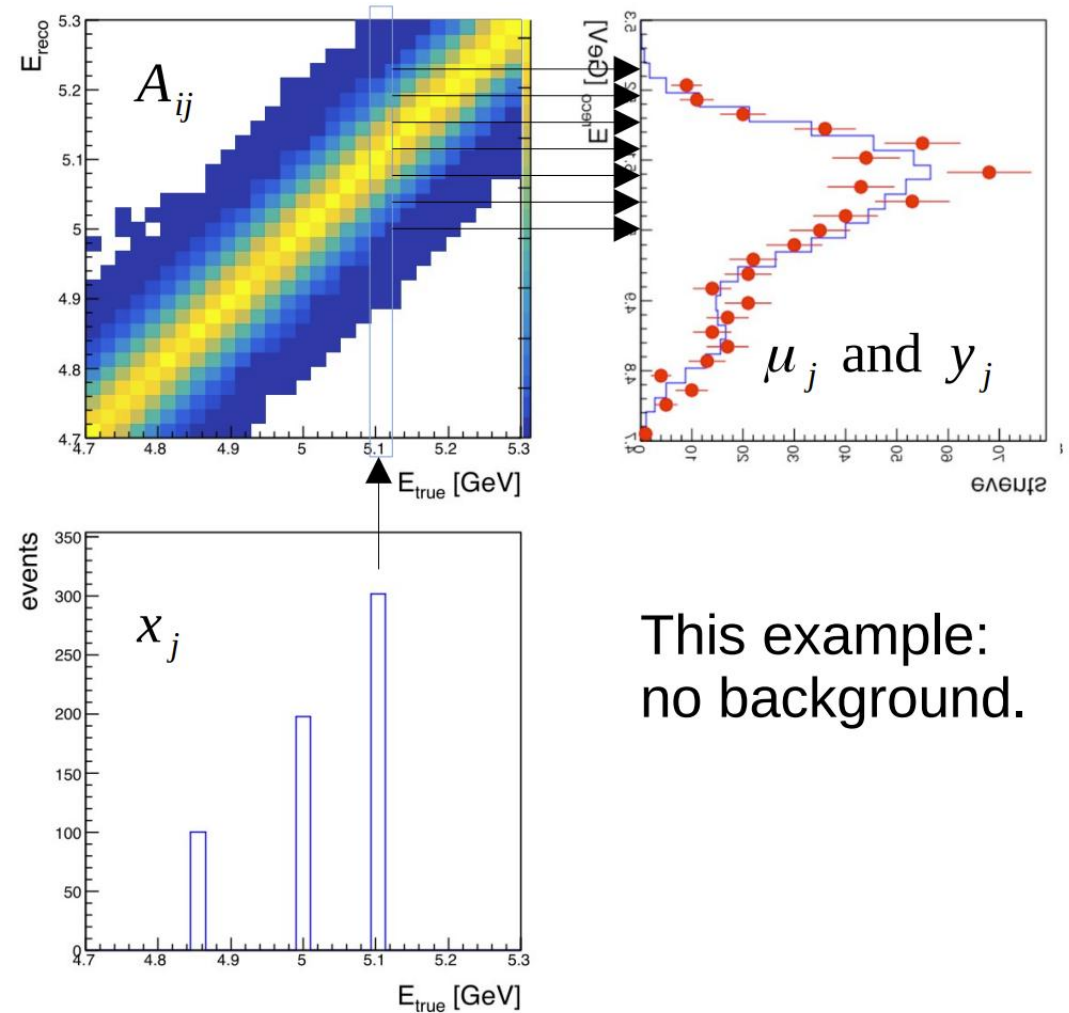
The diagram shows the equation $n(E_i) = \sum_j R_{ij} \cdot n(C_j) + b_i$. A red arrow points from 'Effect (Measurement)' to $n(E_i)$. A yellow arrow points from 'Detector Response Matrix' to R_{ij} . A blue arrow points from 'Cause (Truth)' to $n(C_j)$. The term b_i is enclosed in a grey box with the label 'Background' to its right.

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This example:
no background.

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Effect

(Measurement)

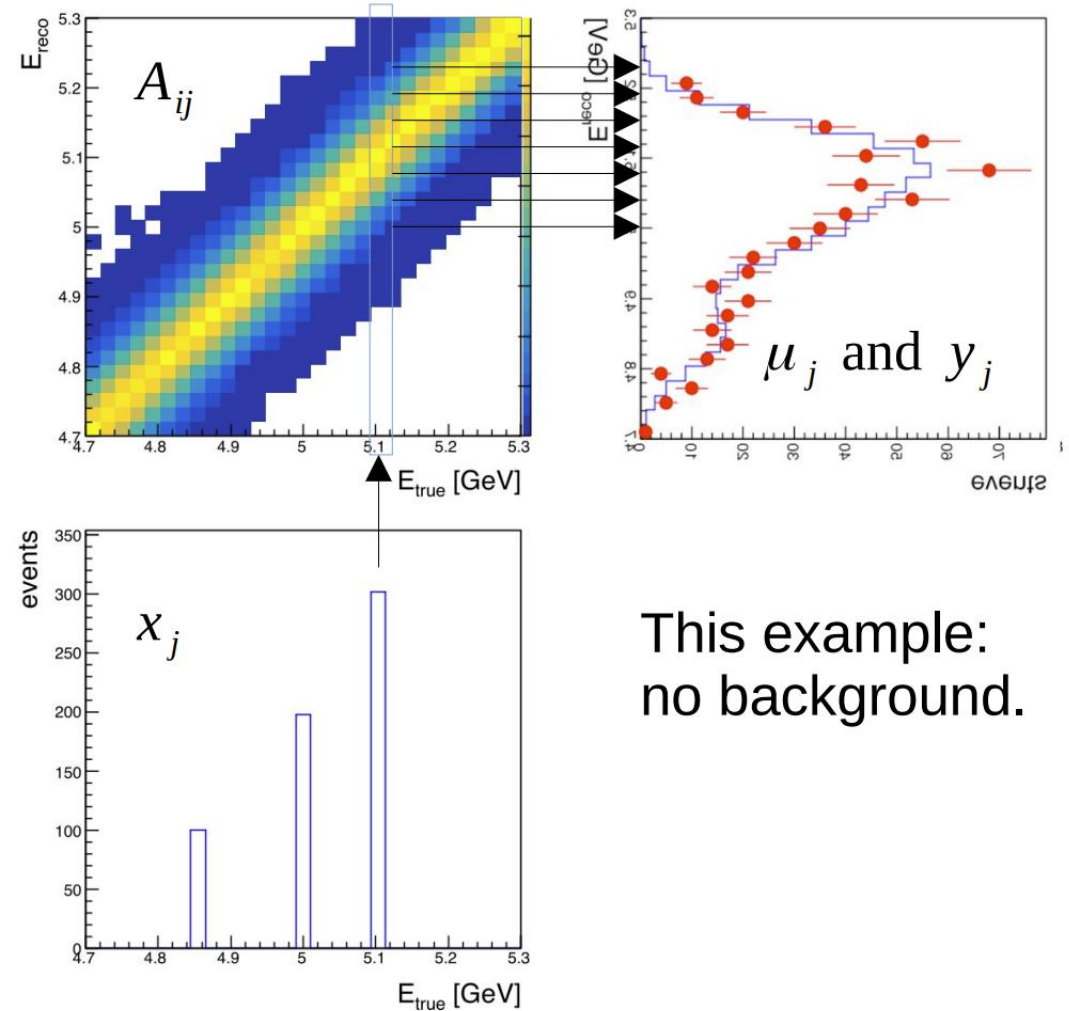
Detector

Response Matrix

Cause

(Truth)

- Response matrix represents the probability of measuring an effect given a cause
- How can we get from a measured spectrum to the most probable truth spectrum



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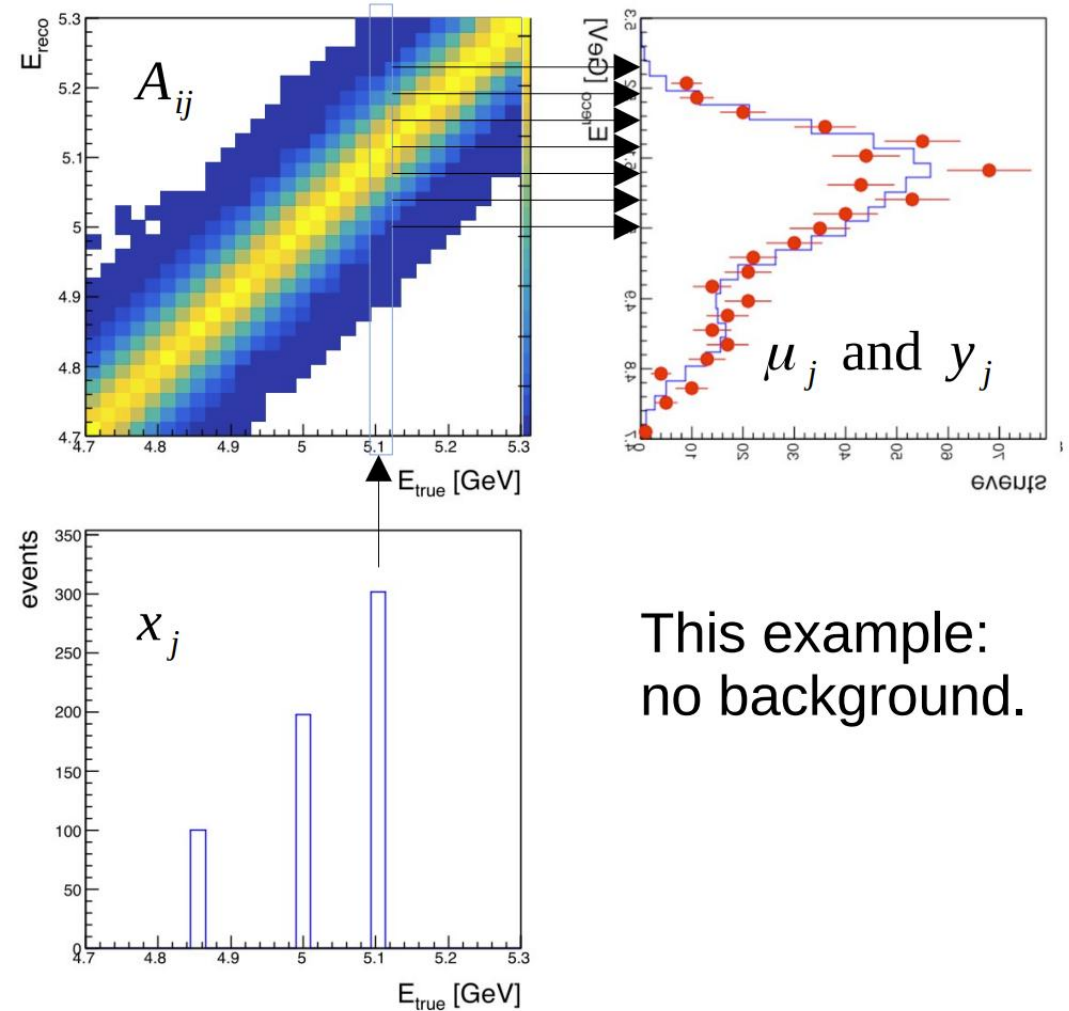
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→ First need response matrix



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 - Resulting „measured“ spectra per cause (energy deposition)

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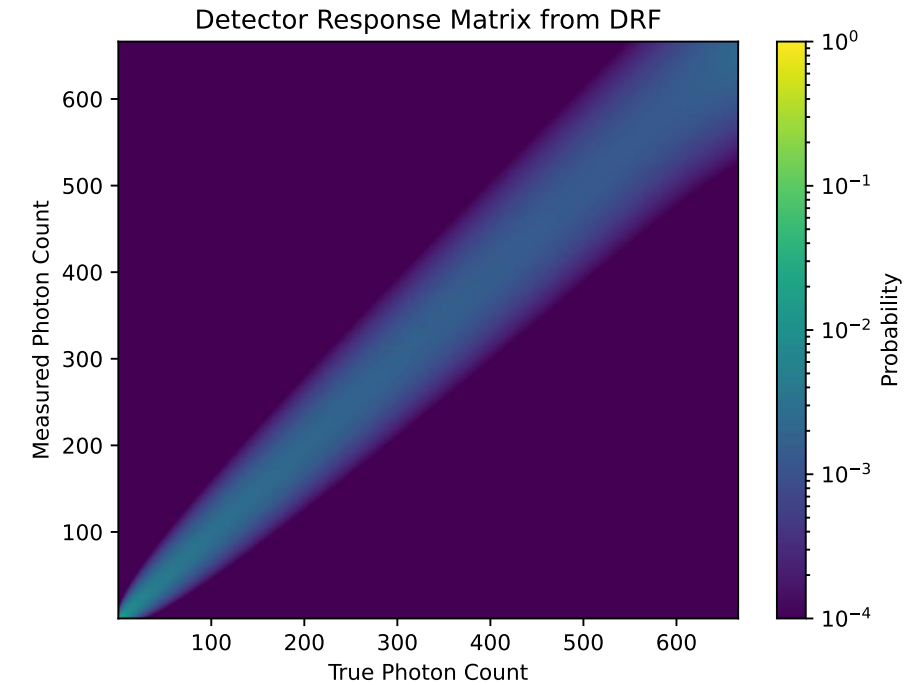
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Response Matrix II

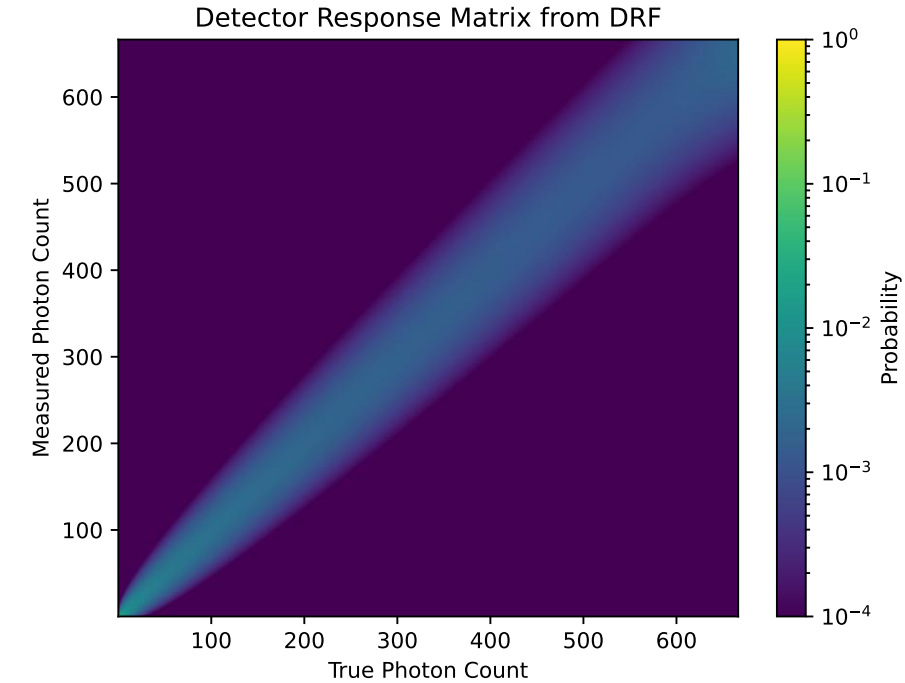
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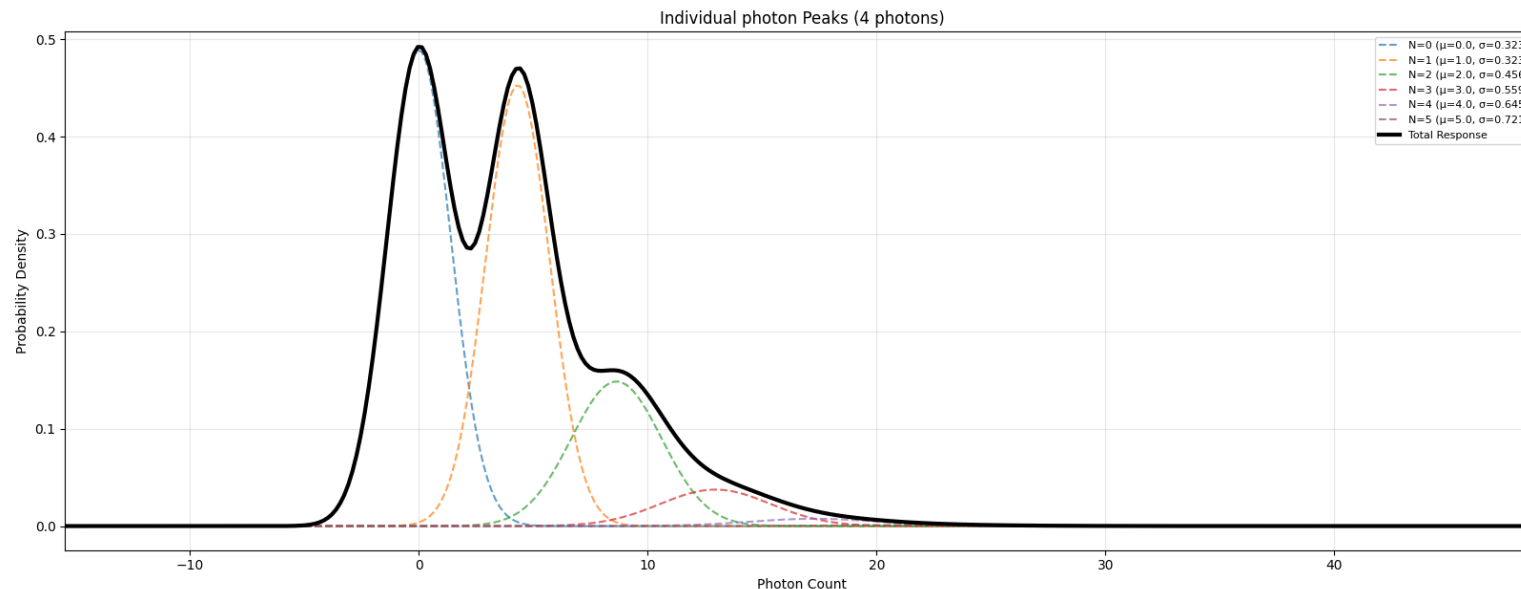
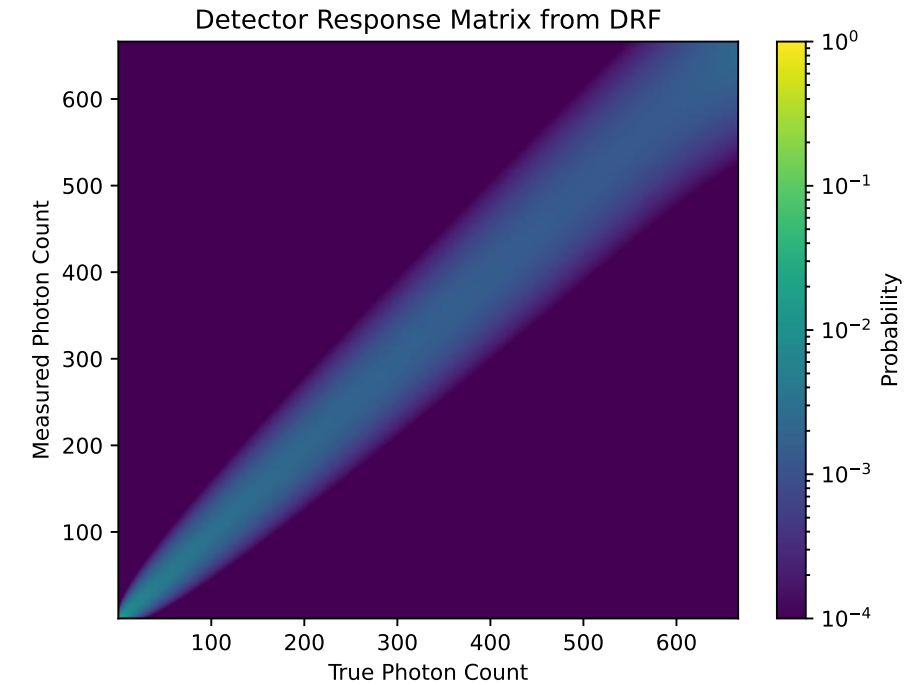
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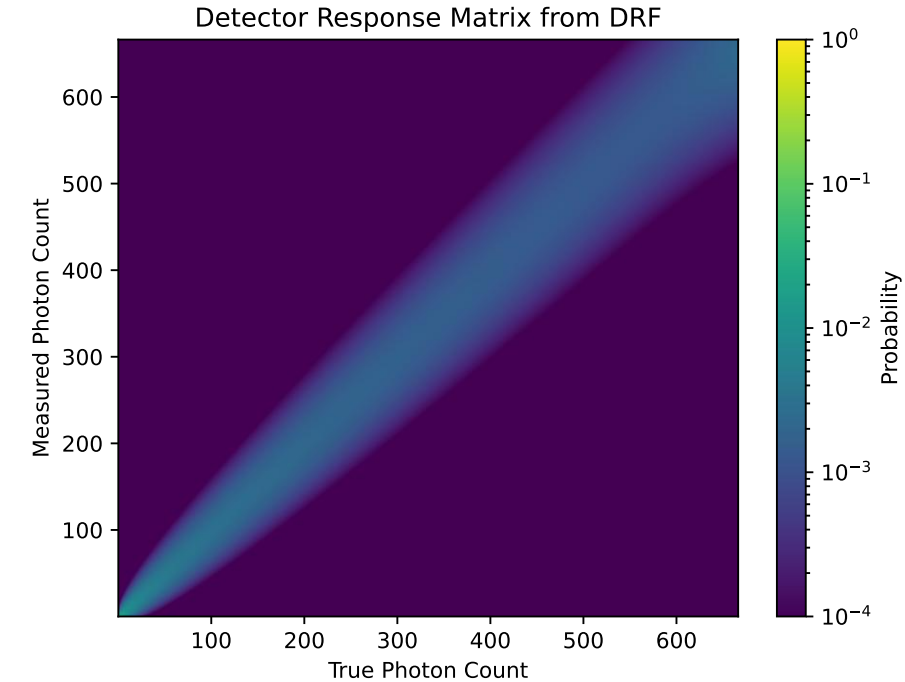
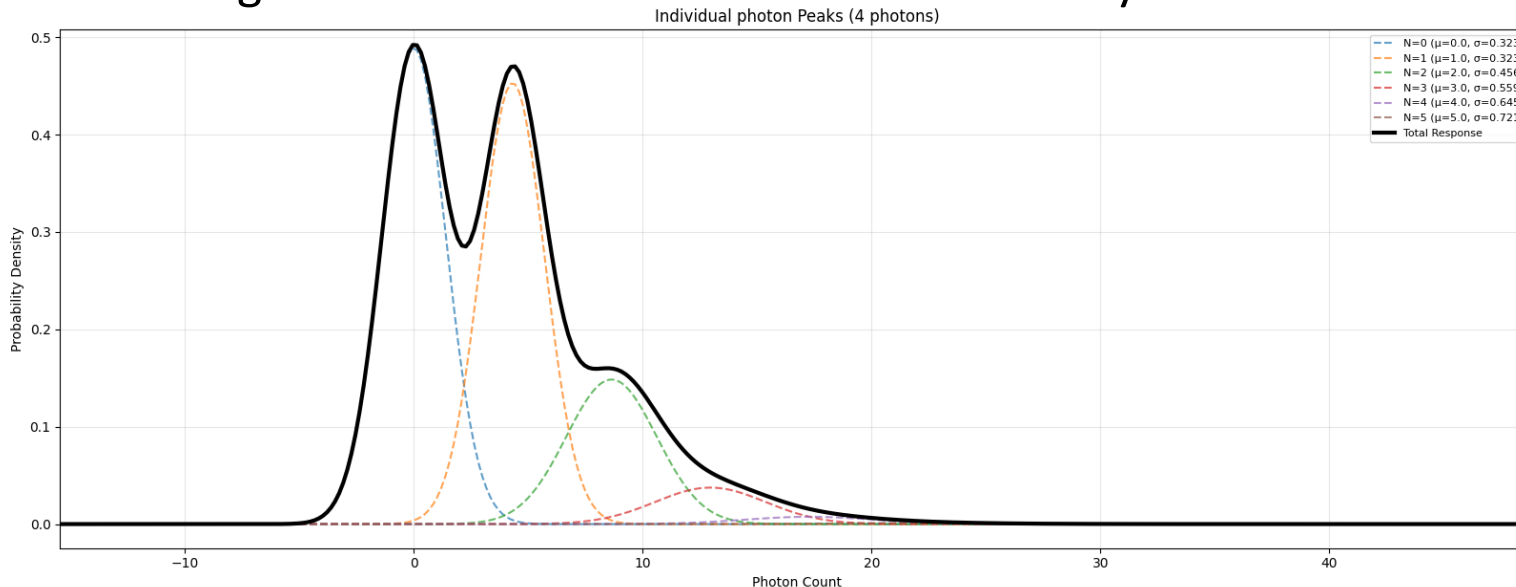
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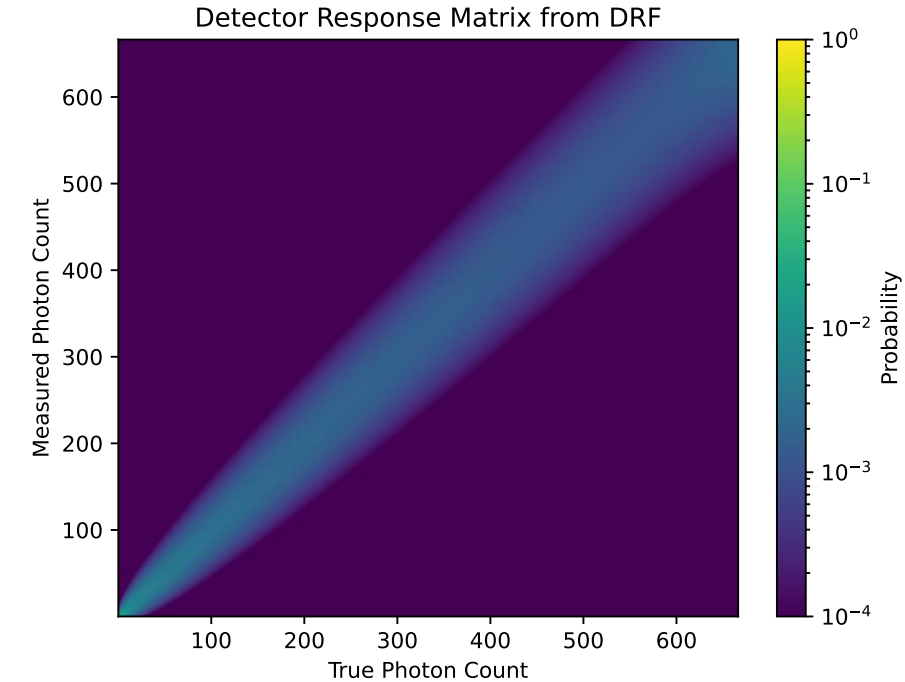
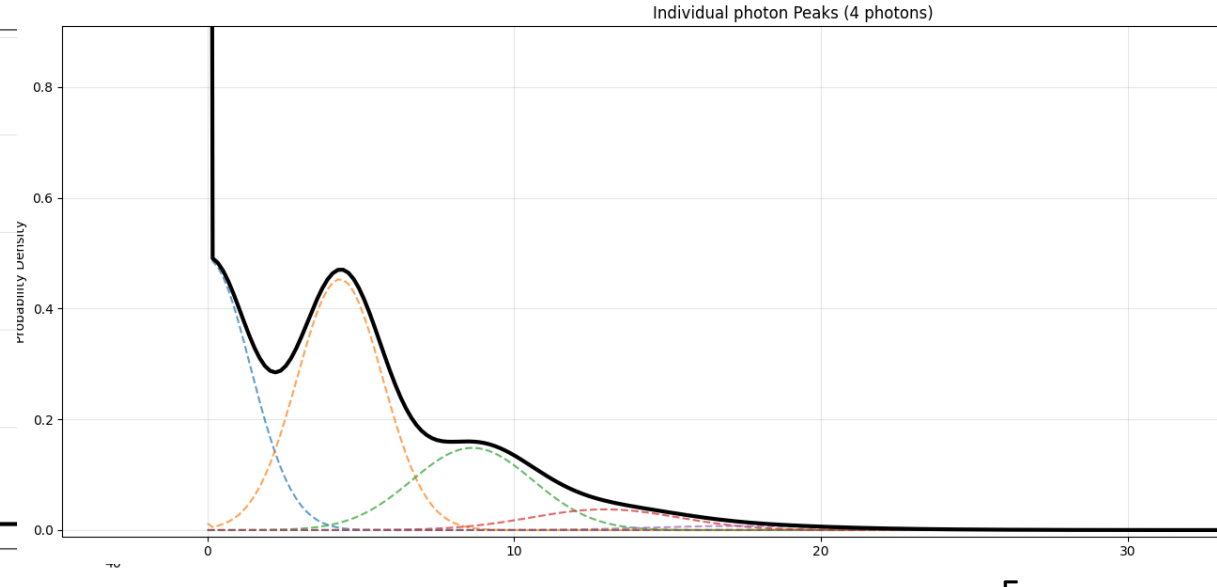
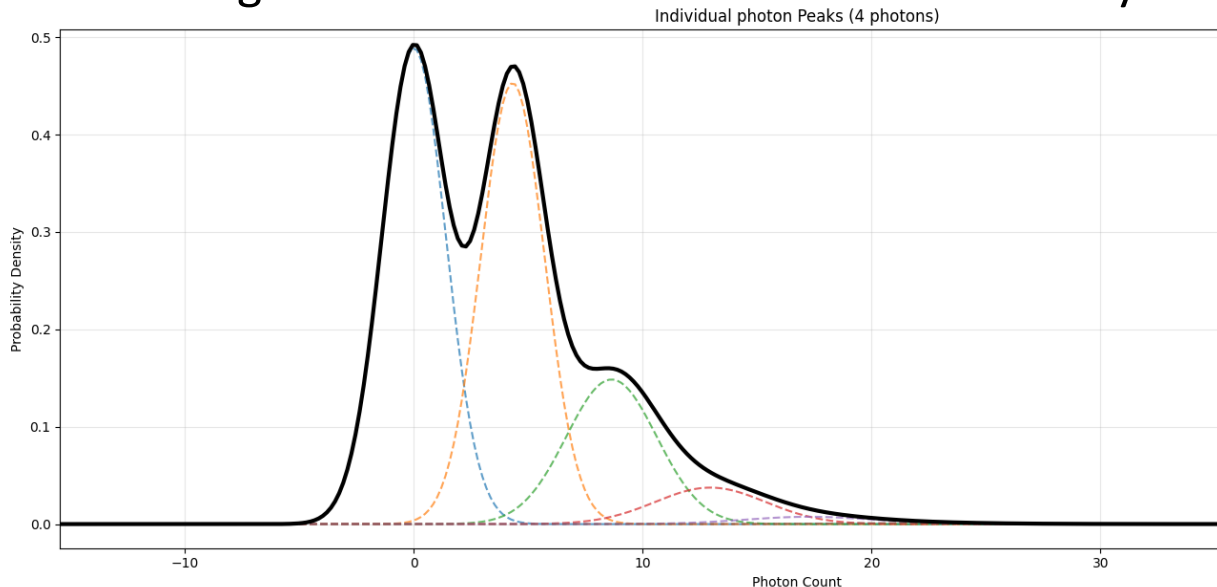
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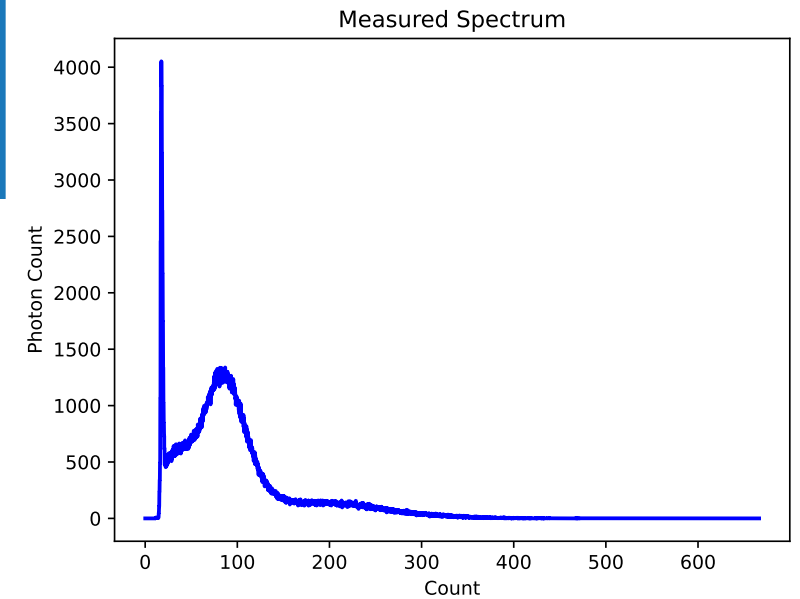
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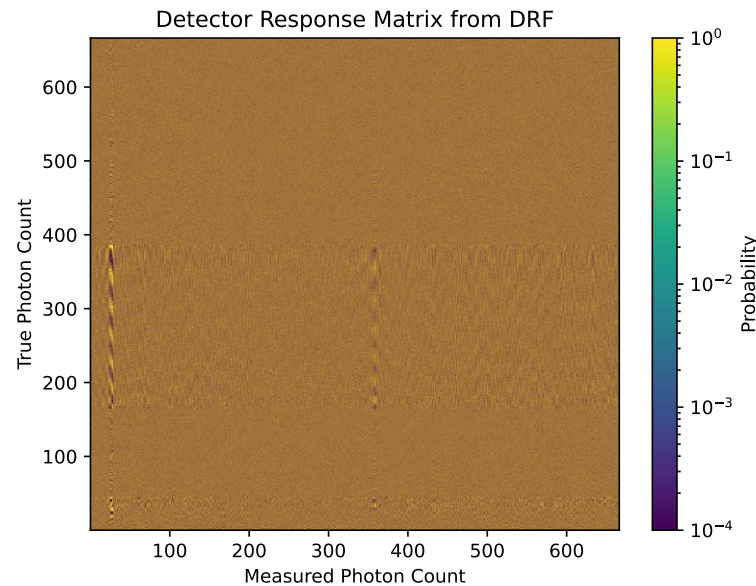
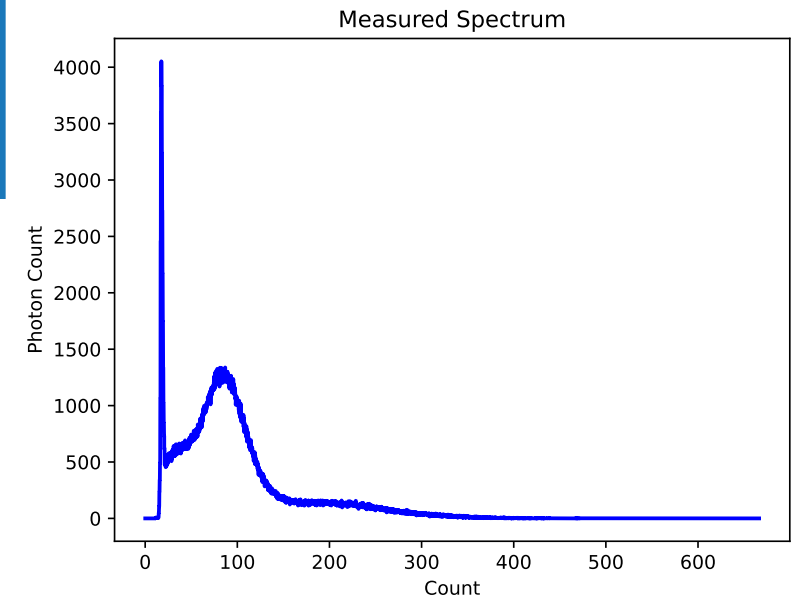


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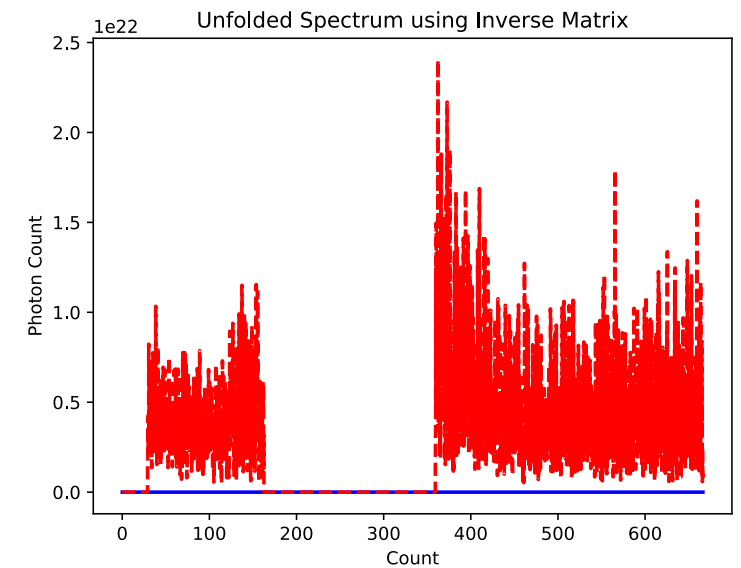
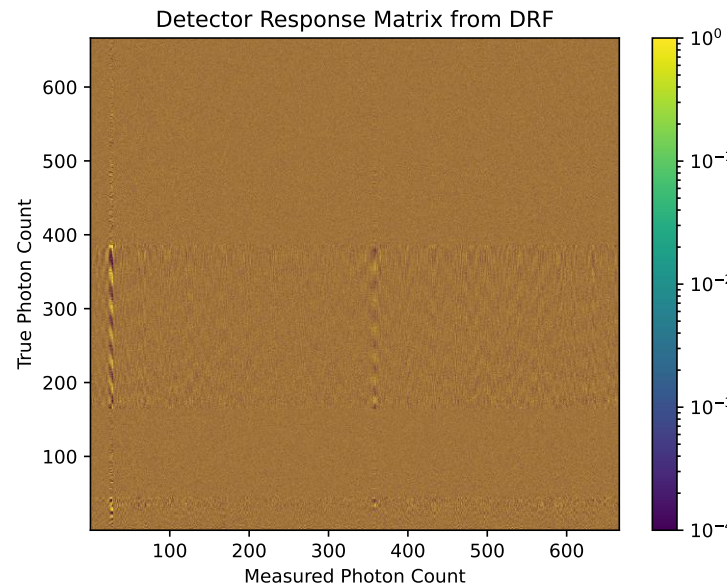
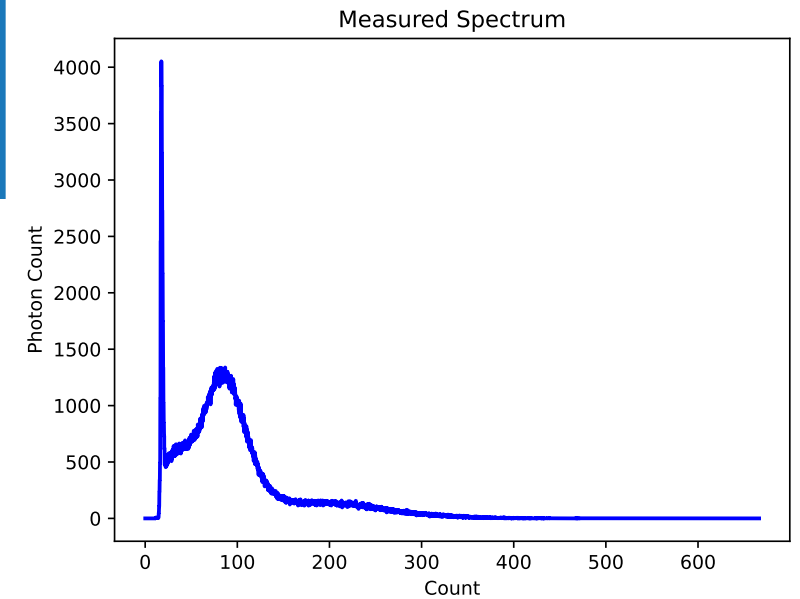


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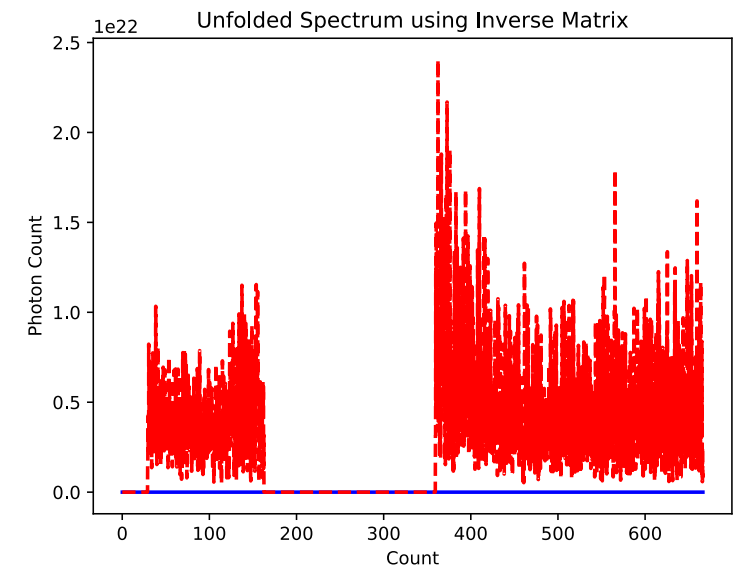
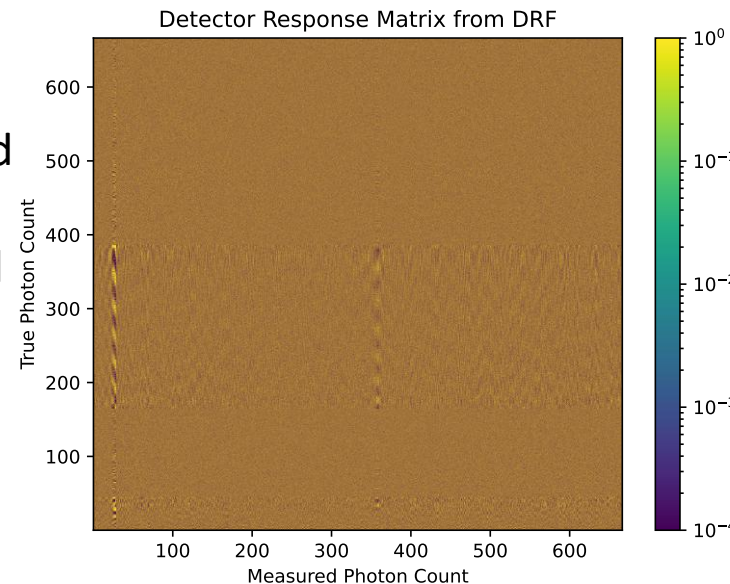
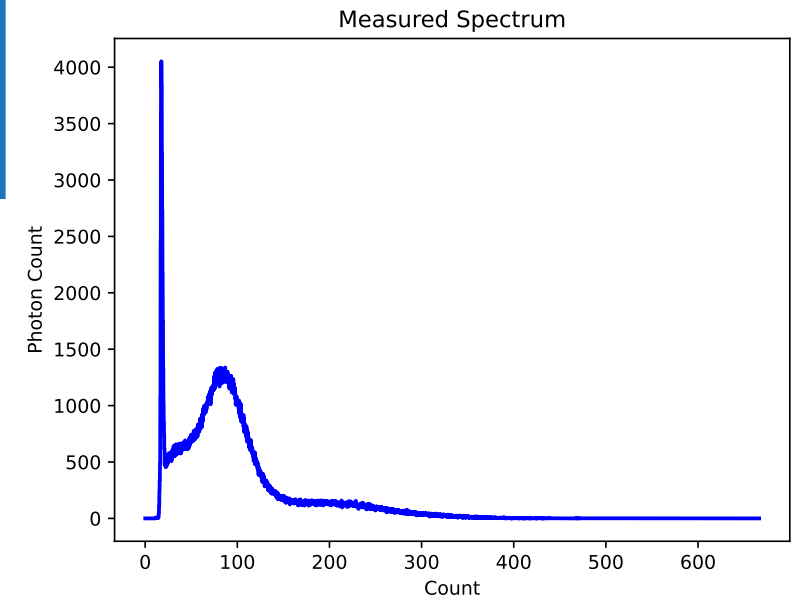


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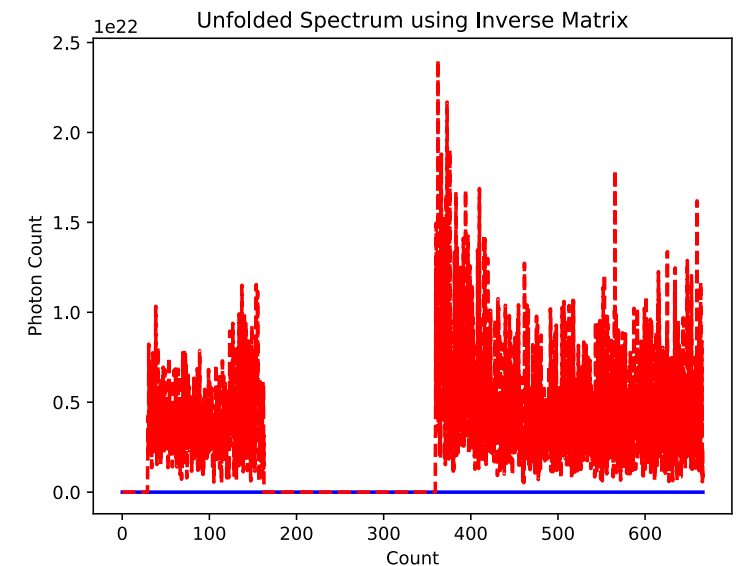
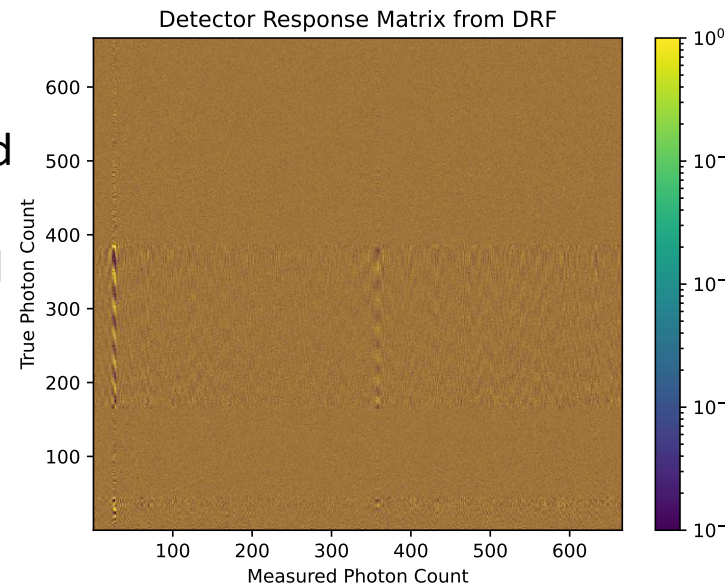
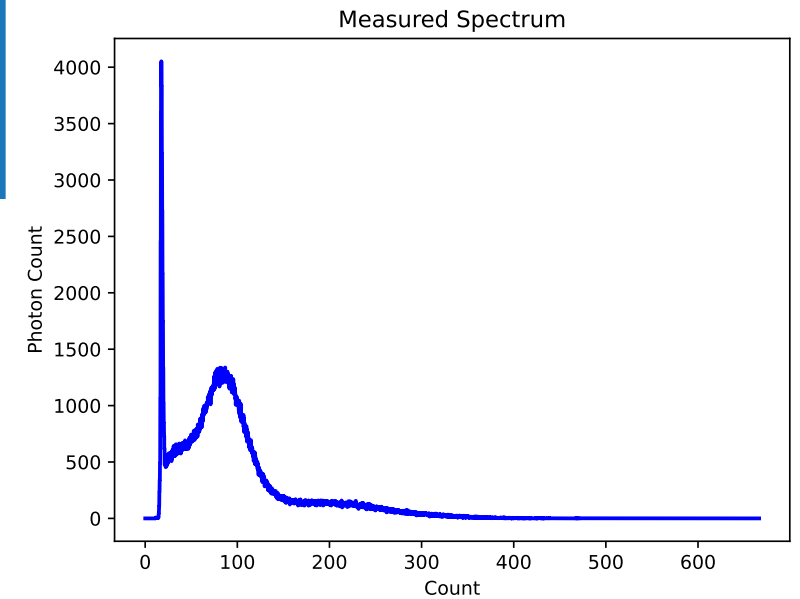


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- Initial guess for $P_0(C_j)$ as flat or just the measurement $n(E_i)/N = P_0(C_i)$

$$n_m(C_j) = \frac{1}{\sum_j R_{ij}} \cdot \frac{R_{ij}^T \cdot n(E_i) \cdot n_{m-1}(C_j)}{R_{ij} \cdot n_{m-1}(C_j)}$$

Unfolding: Bayesian Method (G. D'Agostini 1995)

- Cause-Effect formalism using Bayes theorem: Measurement $n(E)$; Truth $n(C)$
- Bayes' theorem: $P(C_j|E_i) = \frac{P(E_i|C_j) \cdot P_0(C_j)}{\sum_k P(E_i|C_k) \cdot P_0(C_k)}$ (represents unfolding matrix)
- Response matrix as conditional probability $P(E_i|C_j)$

- Calculate true spectrum

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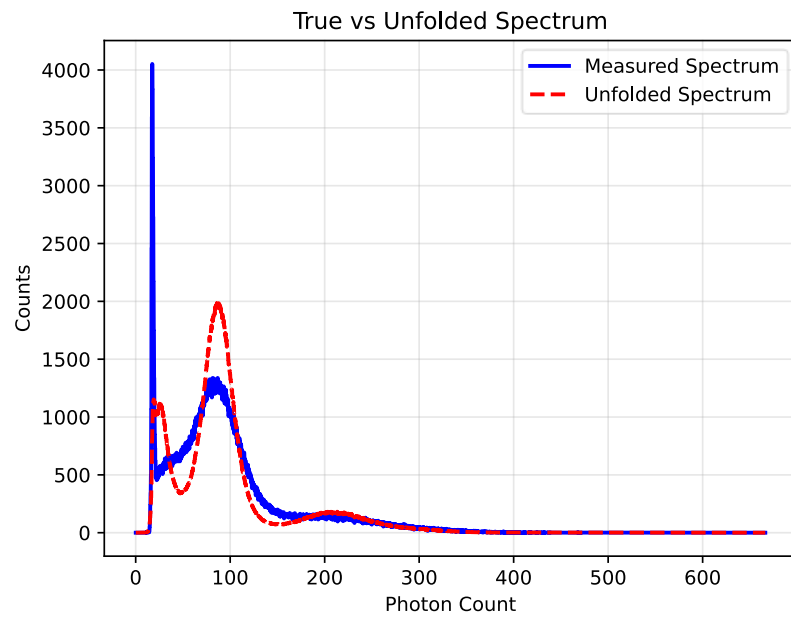
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 - Helps against oscillations

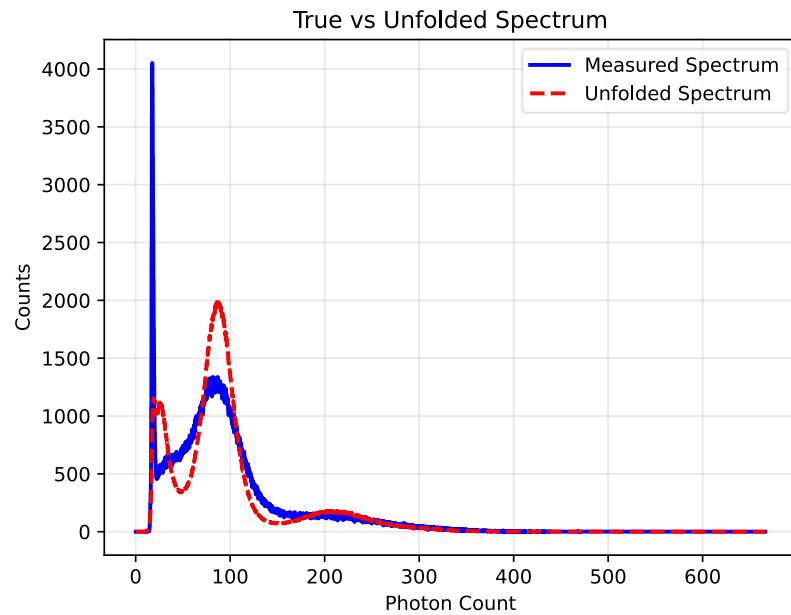
Unfolding Results

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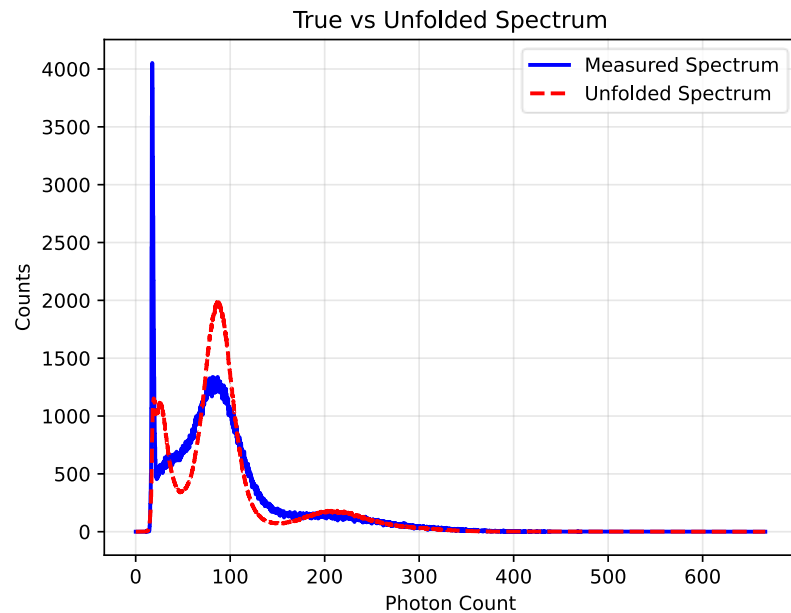
Unfolding Results

- Unfolding improved energy resolution & resolved 1.275 keV Na-22 line



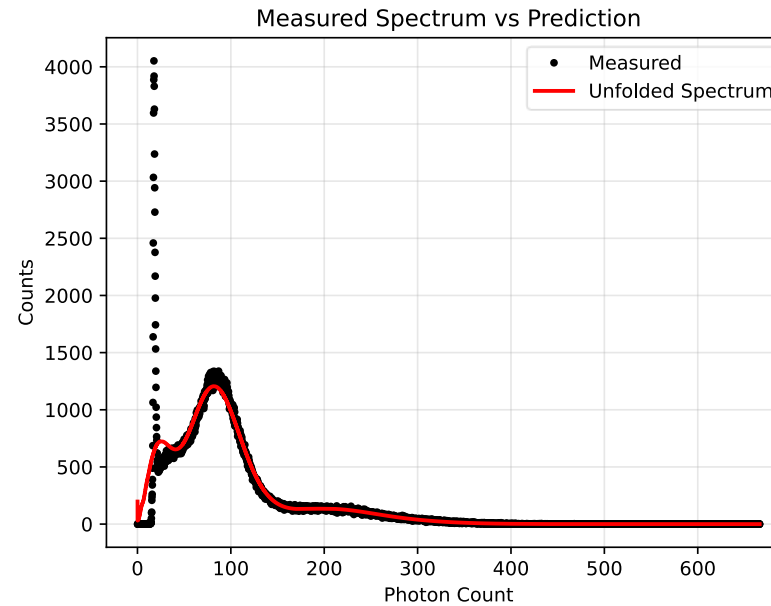
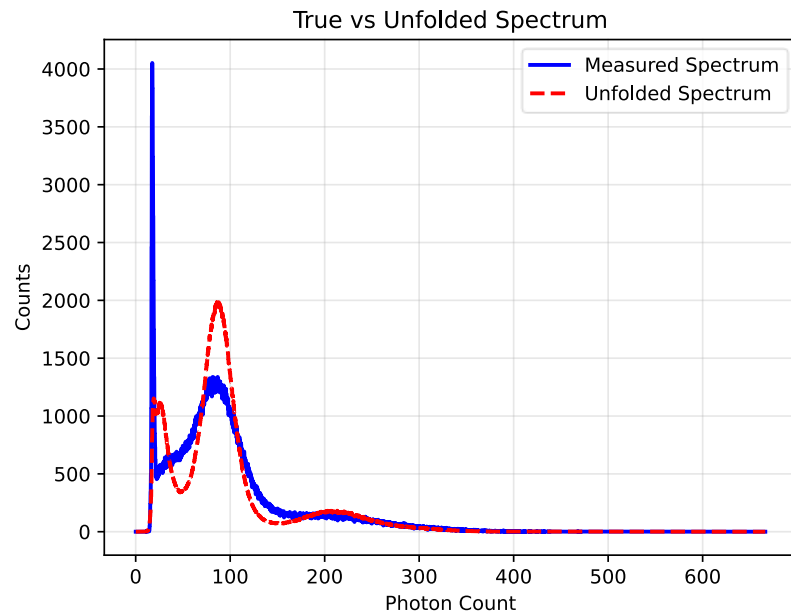
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